

Big Data &amp; SQL

INSTANT

Interactive Dashboards

Data Analysis Diploma • Session 01

# DATA ANALYSIS FUNDAMENTALS

Your Complete Roadmap to Become a Job-Ready Data Analyst

**8 Stages**

Curriculum Roadmap  
From zero to job-ready

**7 Modules**

Applied Core Modules  
Real-world projects

**5 Core Tools**

Industry Tech Stack  
Excel • SQL • Python • BI •  
Tableau

Predictive Analytics

Start Session 01 →

Python &amp; Statistics

— Definition & The 5 Core Activities

# What is Data Analysis?

The process of turning raw data into decisions that matter.



## INDUSTRY DEFINITION

Data Analysis is the process of collecting, cleaning, transforming, analyzing, and interpreting data to discover useful insights and support decision-making.

## KEY ACTIVITIES



### Collect Data

Gather raw data from multiple sources: files, databases, APIs, and forms.

Stage 01



### Clean Data

Fix errors, remove duplicate records, and handle missing null values.

Stage 02



### Analyze Data

Apply statistics, logic, and models to find patterns and trends.

Stage 03



### Visualize Results

Turn complex numbers into clear, readable charts and interactive dashboards.

Stage 04

— Business Impact & Value Creation

# Why Data Analysis is Important?

Every strong business decision starts with reliable data.



## Better Decision Making

Replace guesswork and gut feelings with evidence-based strategic choices.



## Understanding Customers

Reveal behavior patterns, purchasing journeys, and product preferences.



## Finding Opportunities

Spot untapped market niches and uncover hidden business gaps before competitors.



## Improving Performance

Track key performance indicators (KPIs) to optimize operational efficiency.

— Everyday Applications Across Industries

# Data in Real Life

Data is everywhere — every industry runs on it.



## Business

Sales transactions, customer churn, financial records, and supply chain logistics.

• System Standard



## Healthcare

Patient electronic health records, treatment efficacy, and hospital capacity planning.

• System Standard



## Technology

User clickstream interactions, server logs, API telemetry, and app retention metrics.

• System Standard



## Social Media

Posts, comments, sentiment analysis, engagement rates, and viral content reach.

• System Standard

— The Value Hierarchy from Raw Facts to Action

# Data vs Information vs Insight

Three stages of the same raw material, each more valuable than the last.



## 1. Data

Level 1 • Raw Facts

Unprocessed facts, figures, and raw measurements with no context on their own.

EX: "Raw sales numbers: 1400, 2900, 5100"



## 2. Information

Level 2 • Structured Context

Data organized into a structured, readable report with clear context and labels.

EX: "Monthly sales report: Branch Cairo total revenue is \$150,000"



## 3. Insight

Level 3 • Actionable Value

The "why" behind the numbers — actionable business meaning that steers decision-making.

EX: "Sales increased 25% due to the targeted social media promotional campaign"

## The Two Fundamental Data Categories

# Types of Data

Every dataset is built from these two fundamental categories.

DATA CLASSIFICATION SYSTEM:

All Types

Qualitative Data

Quantitative Data



CATEGORICAL • QUALITATIVE

## Qualitative Data

Non-Numerical

Descriptive characteristics, labels, and groups that cannot be mathematically calculated directly.

### NOMINAL (NO ORDER)

Categories with no intrinsic order

Cities (London, NY)

Colors (Red, Blue)

Gender (M/F)

### ORDINAL (ORDERED)

Categories with meaningful rank

Rank (Low, Med, High)

Rating (1 to 5 Stars)

Education Level

Target Visuals: Bar Charts, Donut, Frequency

Categorical



NUMERICAL • QUANTITATIVE

## Quantitative Data

Calculable Numbers

Measurable numerical values that support full mathematical calculations and statistical aggregations.

### DISCRETE (COUNTABLE)

Countable whole integers

Order Count (42)

Employees (150)

Page Visits (8)

### CONTINUOUS (MEASURABLE)

Measurable infinite decimals

Revenue (\$1,249.50)

Duration (4.25 min)

Target Visuals: Histograms, Line, Scatter, KPIs

Numerical

## Data Architecture & Schema Organization

# Structured vs Unstructured Data

How organized is the data before it reaches an analyst?



TABULAR / SQL

## STRUCTURED

Highly organized data stored in standardized schemas, easily queryable with SQL and Excel.

- SQL Database Tables
- Excel Spreadsheets
- Clean CSV Datasets
- Enterprise ERP Systems

Pillar 01

01



JSON / APIS

## SEMI-STRUCTURED

Data that does not conform to strict tabular schemas but carries self-describing tags and keys.

- JSON Web API Payloads
- XML Documents
- Web Server Access Logs
- NoSQL Collections (MongoDB)

Pillar 02

02



~80% OF WORLD DATA

## UNSTRUCTURED

Data with no predefined conceptual model, representing ~80% of all generated enterprise data.

- Images & Graphic Files
- Video & Audio Streams
- Emails & Customer Chats
- PDF Reports & Documents

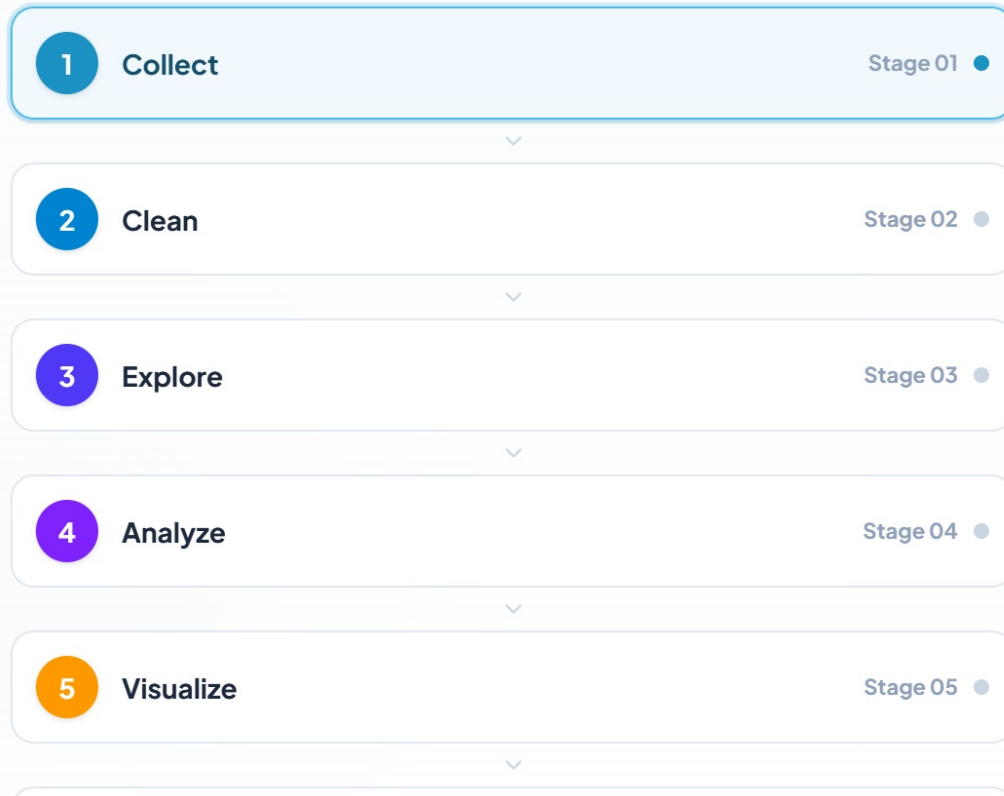
Pillar 03

03

## The 6 Sequential Stages of Every Analytics Project

# Data Analysis Lifecycle

Every analysis project moves through the same six stages.



### STAGE 01 DEEP DIVE

## Collect

Identify and extract raw data from internal and external sources: databases, files, APIs, and forms.

### KEY OUTPUTS & DELIVERABLES:

✓ Identify Sources

✓ Raw Ingestion

✓ Access Verification

Click any stage on the left to inspect

1 / 6

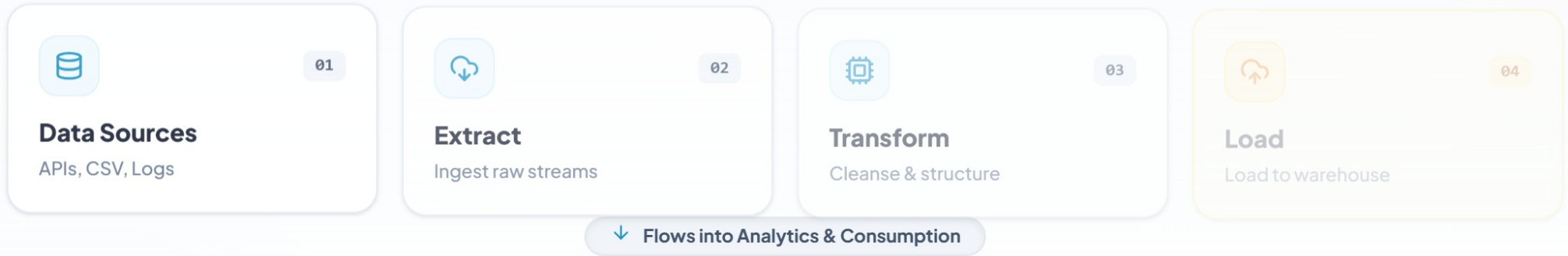
— End-to-End Architecture from Ingestion to Decision

# Data Pipeline

How raw data travels from source to decision.

TIER 1: DATA INGESTION & ETL PIPELINE

ETL PROCESS



TIER 2: STORAGE, ANALYTICS & DECISION

BUSINESS VALUE

## Professional Learning Roadmap

8 Connected Milestones from Zero to Job-Ready

&lt; 01 / 08 &gt;



Math &amp; Stats



Excel



Python Core



Pandas / EDA



SQL / DB



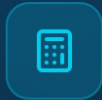
Power BI



Tableau



Web Scraping



STAGE 01 • SCIENTIFIC FOUNDATION

# Math & Statistics

Milestone 01 / 08

## OBJECTIVE & CORE FOCUS

Master statistical distributions, probability theory, hypothesis testing, and variance analysis to uncover the mathematical truth behind data.



HANDS-ON DELIVERABLE:

**Statistical evaluation report assessing data health and business hypotheses.**

## KEY COMPETENCIES & TOOLS

• Descriptive Statistics

• Probability &amp; Distributions

• Hypothesis Testing (A/B)

• Outlier Detection

SECTION 01

# Excel for Data Analysis

The foundation every data analyst starts with.



PART 001

— Core Strengths of Excel in Analytics

# Why Excel is Important

It's the most accessible tool to start organizing and exploring data.



## Data Entry & Organization

Structure raw information into clean rows and columns.



## Formulas & Functions

Automate calculations across large datasets with precision.



## Data Cleaning

Remove duplicates, fix errors, and standardize formats.



## Pivot Tables

Summarize and cross-tabulate large datasets instantly.

— Everyday Analytical Functions

# Formulas & Functions in Practice

A handful of functions cover most everyday analysis needs.

## SUM

Calculate Total Sum

fx

## AVERAGE

Compute Arithmetic Mean

fx

## IF

Conditional Logic & Tiering

fx

## XLOOKUP

Modern Search & Relation...

fx

● Sheet1

fx =XLOOKUP("ORD-104", A2:A6, B2:B6)

| # | A (Order ID) | B (Customer)  | C (Sales Revenue) | D (Loyalty Tier) |
|---|--------------|---------------|-------------------|------------------|
| 2 | ORD-101      | Sarah Connor  | \$1,200           | Regular          |
| 3 | ORD-102      | Alex Mercer   | \$2,450           | VIP              |
| 4 | ORD-103      | Elena Rostova | \$850             | Regular          |
| 5 | ORD-104      | Michael Scott | \$2,150           | VIP              |
| 6 | ORD-105      | David Kim     | \$1,200           | Regular          |

### ● LIVE DYNAMIC OUTPUT

## Modern Search & Relational Lookup

Modern flexible replacement for VLOOKUP without left/right lookup limits.

Lookup Return Value

Michael Scott



Pivot Tables aggregate and group these calculations across millions of rows with drag & drop!

## End-to-End Excel Pipeline

# From Clean Data to Dashboards

The final step: turning a clean table into a visual story.

01

### Data Cleaning & Prep

Active Stage

02

### Build Pivot Tables

Click to inspect

03

### Insert Pivot Charts

Click to inspect

04

### Connect Slicers & Timeline

Click to inspect

STAGE 01 / 04 STEP 1

## Data Cleaning & Prep

Remove duplicates, handle blanks, and format ranges into official Excel Tables (Ctrl+T).

- ✓ Convert data range into Table (Ctrl+T)
- ✓ Remove duplicate entries automatically
- ✓ Standardize dates, currencies, and numbers

Milestone Target:

100% Cleaned Data



Excel Workspace Simulation

Live Flow

Raw Records

15,420 rows

Nulls Cleaned

824 fixed

Ready Dataset

14,596 rows

Previous Step

1 of 4

Next Step

## SECTION 02

# Python Fundamentals

The programming language that powers modern data analysis.



PART 002

— Core Building Blocks in Code

# Variables & Data Types

Python's role: automating analysis beyond what Excel can handle.



## Variables & Fundamental Data Types

The core building blocks for storing and processing data in Python

- int / float: Store integer and decimal numbers (prices, quantities, metrics)
- str: Store text strings (customer names, categories, and geographic locations)
- bool: Store logical True/False flags for boolean conditions and filters
- type(): Built-in Python function to inspect and validate variable data types

Interactive Code Implementation

PYTHON

PYTHON SCRIPT

Copy Code

```
# Customer Transaction Record
customer_name = "Sarah Johnson"      # str
order_id = 10482                      # int
total_amount = 249.75                # float
is_loyalty_member = True              # bool

# Compute loyalty discount dynamically
discount = 0.15 if is_loyalty_member else 0.0
final_price = total_amount * (1 - discount)

print(f"Customer: {customer_name}")
print(f"Order #{order_id} Total: ${final_price:.2f}")
```

>\_ Ready to execute in Jupyter / VSCode / SQL Console

● Valid

## — Branching & Iteration

# Conditions & Loops

Logic that lets code react to data and repeat tasks automatically.

PYTHON LOGIC & ITERATION ENGINE:

[Core Pillars \(PDF View\)](#)
[🌟 Interactive Lab](#)


BRANCHING LOGIC

## Conditions

if / elif / else

- if / elif / else evaluate a condition
- Used to filter, flag, or branch logic
- Example: flag rows where revenue < target

```
if row['revenue'] < target:
    row['flag'] = 'Under Target'
```



ITERATIVE LOGIC

## Loops

for / while

- for & while loops repeat code block
- Iterate over datasets row by row
- Example: process all customer records

```
for client in dataset:
    process_record(client)
```

Conditions and loops form the fundamental pillars of data pipeline automation in Python

## Reusable Data Structures & Logic

# Functions, Lists & Dictionaries

Structures that organize data and logic for reuse.



## Functions, Lists & Dictionaries

Data structures and reusable modular logic for automation

- Lists []: Ordered arrays to store metric sequences (e.g. daily sales records)
- Dictionaries {}: Key-value pairs representing structured entity profiles
- Functions def: Modular functions to automate repetitive analytical formulas
- Loops for / while: Iterate through datasets to clean and transform records

PYTHON SCRIPT

Copy Code

```
# 1. List of Daily Sales Totals
daily_sales = [1200, 1850, 2100, 950, 3200]

# 2. Dictionary of Customer Entity
customer = {
    "id": "C-901",
    "name": "Ahmed Hassan",
    "city": "Cairo",
    "orders_count": 12
}

# 3. Reusable KPI Calculation Function
def calculate_growth(current, previous):
    rate = ((current - previous) / previous) * 100
    return round(rate, 2)
```

## SECTION 03

# Python for Data Analysis

Putting Python libraries to work on real datasets.



PART 003

## The Two Indispensable Libraries

# NumPy & Pandas

The two libraries every data analyst uses daily.

PYTHON CORE DATA STACK ECOSYSTEM:

Dual Comparison

NumPy Engine

Pandas DataFrames



NUMERICAL FOUNDATION

## NUMPY

```
import numpy as np
```

- Fast numerical operations
- Works with arrays and matrices
- Foundation for other data libraries
- Powers statistical calculations

N-Dimensional ndarray

[Explore Engine →](#)



TABULAR MANIPULATION

## PANDAS

```
import pandas as pd
```

- Data manipulation and analysis
- DataFrames organize rows and columns
- Filter, group, and merge datasets easily
- Reads CSV, Excel, SQL and more

2D Tabular DataFrames

[Explore DataFrames →](#)

NumPy delivers high-speed numerical array crunching, while Pandas provides flexible tabular DataFrames

## Python Visualizations

# Matplotlib & Seaborn

Python's visualization toolkit for charts and statistical plots.

PYTHON'S VISUALIZATION TOOLKIT:

Dual Comparison

Matplotlib Engine

Seaborn Statistical Lab



BASE PLOTTING ENGINE

## Matplotlib

```
import matplotlib.pyplot as plt
```

The base plotting library — full control over every chart element: line plots, bar charts, histograms.

- Fine-grained control over axes, ticks, colors, labels
- Standard charts: Line, Bar, Histograms, Scatter
- Object-oriented Figure and Subplots architecture

PLOTS: **Line** Bar Hist

[Interactive Demo →](#)



STATISTICAL VISUALS

## Seaborn

```
import seaborn as sns
```

Built on Matplotlib for statistical visualization — cleaner defaults for distributions, correlations, and comparisons.

- Correlation heatmaps and multi-variable matrices
- Kernel Density Estimation (KDE) and distribution curves
- Native DataFrame integration with high-level API

STATS: **KDE** Heatmap Box

[Interactive Demo →](#)

Matplotlib gives you low-level architectural control, while Seaborn delivers effortless statistical beauty



## SECTION 04

# SQL & Databases

The language of structured data — used in nearly every company.



PART 004

## Relational Data Storage

# Why Databases Matter

Where structured business data actually lives.

RELATIONAL DATABASE ARCHITECTURE:

[Core Pillars \(PDF View\)](#)
[Interactive Schema](#)


ACID Safe

STORAGE ENGINE

## Database Concepts

A structured system for storing and retrieving data reliably.

Integrity & Concurrency



Rows & Cols

TABULAR SCHEMA

## Tables

Data organized into rows (records) and columns (fields).

Standardized Schema



PK → FK

RELATIONAL KEYS

## Relationships

Tables connect through shared keys, e.g. customer ID.

Primary & Foreign Keys

[View Link](#)

Databases are where enterprise business data lives — relational keys are what give them power

## — Querying & Slicing Tables

# SQL Queries & Filtering

Ask precise questions of your data using SQL.



### Core SQL Querying Architecture

Extracting, filtering, and ordering tabular records from RDBMS

- **SELECT / FROM:** Specify columns to retrieve and the primary source table
- **WHERE:** Filter records using conditional criteria (amount > 1000, active status)
- **GROUP BY / HAVING:** Aggregate rows and apply aggregate thresholds (SUM, COUNT)
- **ORDER BY / LIMIT:** Sort records ascending/descending and isolate top performers

Interactive Code Implementation

SQL



SQL SCRIPT

Copy Code

```
-- Top 5 High-Value Customers in 2024
SELECT
    c.customer_id,
    c.customer_name,
    c.city,
    COUNT(o.order_id) AS total_orders,
    SUM(o.order_amount) AS total_spent
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_date >= '2024-01-01'
GROUP BY c.customer_id, c.customer_name, c.city
HAVING SUM(o.order_amount) > 5000
ORDER BY total_spent DESC
LIMIT 5;
```

## Summaries & Multi-Table Joins

# Aggregation & Joins

Summarize data and combine multiple tables together.



### Relational Joins (INNER, LEFT, FULL)

Combining related records across tables using Primary & Foreign Keys

- INNER JOIN: Returns only rows with matching foreign keys in both tables
- LEFT JOIN: Keeps all rows from left table, with NULLs for unmatched right rows
- RIGHT JOIN: Keeps all rows from right table with corresponding left matches
- FULL OUTER JOIN: Combines all records from both tables regardless of match

Interactive Code Implementation

SQL



SQL SCRIPT

Copy Code

```
-- Left Join to Find Inactive / Non-Purchasing Customers
SELECT
    c.customer_id,
    c.customer_name,
    c.signup_date,
    o.order_id,
    o.order_amount
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL; -- Identifies registered users with 0 purchases
```

>\_ Ready to execute in Jupyter / VSCode / SQL Console

• Valid

## Nested Queries & Database Engines

# Subqueries & SQL Tools

Nesting queries for advanced logic, and where SQL runs.

## Subqueries, CTEs & Window Functions



Nested queries, Common Table Expressions, and analytical rankings

- Subqueries: Nested SELECT statements to evaluate dynamic scalar thresholds
- CTEs (WITH clause): Reusable named query blocks for clean readability
- DENSE\_RANK() OVER(): Compute rank hierarchies per category partition
- PARTITION BY: Execute windowed aggregations without collapsing row granularity

Interactive Code Implementation

SQL



SQL SCRIPT

Copy Code

```
-- Rank Top Selling Products within Each Category
WITH RankedProducts AS (
    SELECT
        category_name,
        product_name,
        revenue,
        DENSE_RANK() OVER (
            PARTITION BY category_name
            ORDER BY revenue DESC
        ) AS rank_in_category
    FROM sales_records
)
SELECT * FROM RankedProducts
WHERE rank_in_category <= 3;
```

SECTION 05

# Power BI

Turning models into business-ready dashboards.



PART 005

## ETL & Star Schema Modeling

# Power Query & Data Modeling

Power BI role in business intelligence: connect, shape, and model data.

POWER BI DATA PREPARATION & ARCHITECTURE:

Core Pillars (PDF View)

Power Query ETL

Star Schema Model



ETL Engine

DATA INGESTION

## Power Query

Connect to sources and clean/shape data before loading it in.

Applied Steps

View ETL



Star Schema

LOGICAL DESIGN

## Data Modeling

Organize tables into a logical model for analysis.

Facts & Dimensions

View Schema



1 : Many (1:\*)

MODEL CARDINALITY

## Relationships

Connect tables together using shared keys.

✓ Cross-Filtering Active

Power Query cleans and shapes data; Data Modeling builds the engine for lightning-fast business insights

## Power BI Calculation Language

# DAX — Data Analysis Expressions

The formula language behind Power BI calculations.



## DAX Measures & Context Transition in Power BI

Authoring dynamic business KPIs and Time Intelligence calculations

- **CALCULATE():** Core DAX engine function that overrides and manipulates filter context
- **Time Intelligence:** Compare period performance (SAMEPERIODLASTYEAR, YTD)
- **Measures:** Computed dynamically on-the-fly to optimize RAM and speed
- **DIVIDE():** Safe mathematical division preventing division-by-zero crashes

Interactive Code Implementation

DAX



DAX SCRIPT

Copy Code

```
-- 1. Dynamic Year-over-Year Revenue Growth
Total Revenue = SUM(Sales[Revenue])

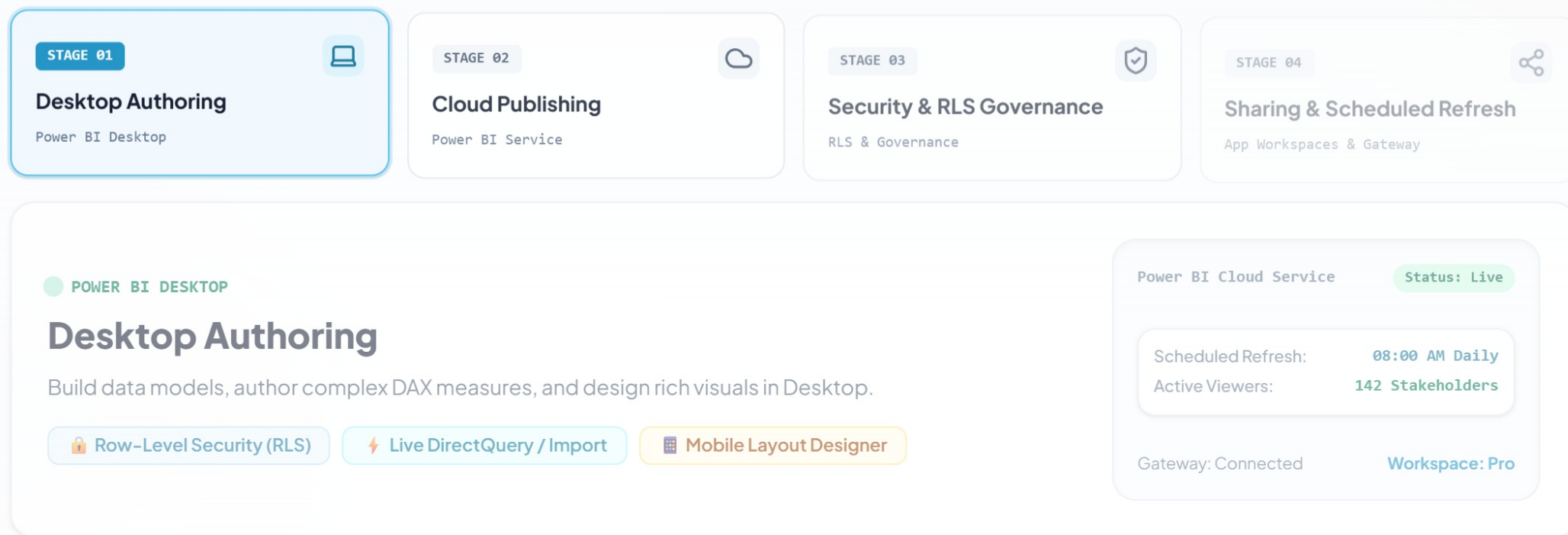
Revenue LY =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR('Calendar'[Date])
)

YoY Growth % =
DIVIDE(
    [Total Revenue] - [Revenue LY],
    [Revenue LY],
    0
)
```

— From Model to Shared Workspace

# Dashboards, Reports & Publishing

From a data model to a shareable business tool.



SECTION 06

# Tableau

A leading tool for visualization and data storytelling.



— Visual Analytics Platform

# Connecting Data & Creating Charts

Tableau as a visualization and business intelligence tool.



COMPONENT 01

## Connecting Data Sources

Plug directly into spreadsheets, relational databases, and cloud data warehouses.

Visual Analytics

01



COMPONENT 02

## Creating Charts

Drag-and-drop chart building with instant interactive visual feedback.

Visual Analytics

02

## Engaging Stakeholder Communication

# Interactive Dashboards & Storytelling

Tableau's real strength: making data explorable and memorable.



COMPONENT 01

## Interactive Dashboards

Filters and actions let viewers explore the data dimensions themselves.

Visual Analytics

01



COMPONENT 02

## Storytelling with Data

Sequence dashboards into a guided narrative for executive stakeholders.

Visual Analytics

02

— BI Industry Giants Comparison

# Power BI vs Tableau

Two leading BI tools — often complementary, not competing.



PILLAR 01

## POWER BI

Tight Microsoft ecosystem integration and powerful DAX modeling.

- Tight Microsoft ecosystem integration
- Strong data modeling with DAX
- Cost-effective licensing
- Great for business reporting

Core Concept

01



PILLAR 02

## TABLEAU

Best-in-class visual flexibility and exploratory drag-and-drop analysis.

- Best-in-class visual flexibility
- Deep, intuitive drag-and-drop analysis
- Popular for storytelling & exploration
- Strong for advanced visual analytics

Core Concept

02

SECTION 07

# Web Scraping

Collecting data directly from the web when no dataset exists.



PART 007

## Automated HTML Extraction

# What is Web Scraping?

Extracting data automatically from websites and web pages.



### HTML Structure

Web pages are built from tags — scraping parses that structural hierarchy.

• System Standard



### Requests

A Python library used to fetch a web page's raw HTML content over HTTP.

• System Standard



### BeautifulSoup

Parses HTML DOM trees so specific elements and text can be extracted.

• System Standard



### APIs

A cleaner, structured alternative to scraping raw HTML pages.

• System Standard

## Real-world Scraping Applications

# Web Scraping in Practice

Real scenarios where analysts use scraping to fill data gaps.

WEB SCRAPING REAL-WORLD SCENARIOS:

[Core Pillars \(PDF View\)](#)
[Price Tracker Lab](#)
[Scheduled Automation](#)


E-Commerce

COMPETITIVE INTEL

## Collecting Product Prices

Track competitor pricing across e-commerce sites.

Dynamic Pricing

[Try Scraper](#)


Public Data

MARKET RESEARCH

## Market Analysis

Aggregate public data to study market trends.

[Trend Aggregation](#)


Cron Job

SCHEDULED SCRIPTS

## Data Collection Automation

Schedule scripts to gather fresh data on a routine.

Cron Pipelines

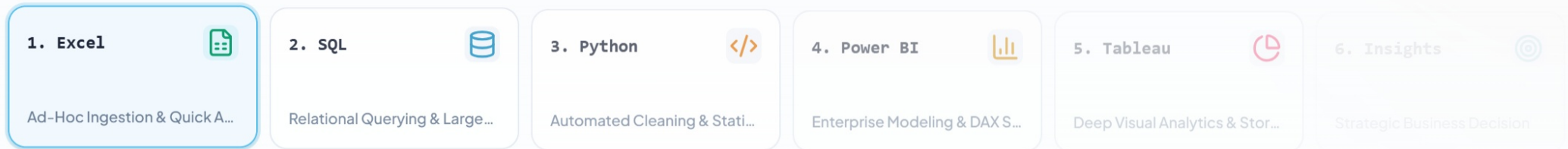
[View Schedule](#)

Web scraping fills critical data voids when no formal dataset or public API exists

## — End-to-End Tool Pipeline

# Complete Data Analyst Workflow

How every tool in this roadmap fits into one pipeline.



STAGE 01 / 06

AD-HOC INGESTION &amp; QUICK AUDITS

Clean Tabular Spreadsheets

## 1. Excel: Ad-Hoc Ingestion & Quick Audits

Ingest raw sheets, perform ad-hoc checks, and run rapid initial summaries via Pivot Tables.

🌟 Harmonious Synergy: Each tool amplifies the next in a modern enterprise stack!

Previous Tool

Next Tool

— The Full Skill Stack

# Career Skills of a Data Analyst

Technical ability and soft skills both matter for the job.

🏆 THE JOB-READY EQUATION (50% TECHNICAL + 50% BUSINESS & SOFT SKILLS):

Complete Matrix

Technical (50%)

Soft Skills (50%)



HARD SKILLS • 50% WEIGHT

## Technical Skillset

Code & Tools

### Excel & Advanced Spreadsheets

Formulas, Pivot Tables, Data Cleansing

Core

### SQL & Database Architecture

Queries, Joins, Aggregation, Subqueries

Top In-Demand

### Python for Data Analysis

NumPy, Pandas, Matplotlib, Seaborn

Automation

### BI Dashboards (Power BI / Tableau)

DAX Modeling, Power Query, Storytelling

Business Facing



SOFT & BUSINESS • 50% WEIGHT

## Business & Soft Skills

Strategy & Value

### Business Acumen

Connecting metrics directly to revenue and business goals

Strategy

### Data Storytelling & Communication

Translating complex findings into clear executive actions

Influence

### Problem Solving & Logic

Breaking down ambiguous business questions logically

Thinking

### Critical Thinking

Questioning assumptions and validating data veracity

Quality

— Complete Session Recap

# Session Summary

Here's everything we covered in this session.



## What is Data Analysis?

Collecting, cleaning, analyzing, and interpreting data.



## Data Types

Qualitative, quantitative, structured, and unstructured.



## Data Pipeline

Sources → Extract → Transform → Load → Analysis.



## Analysis Lifecycle

Collect, clean, explore, analyze, visualize, communicate.

— Interactive Self-Assessment

# Knowledge Check

Test your understanding before moving to the next session.

1 1. What is the difference between structured & unstructured data?

Click to reveal 💡

2 2. What are the 6 steps of the Data Analysis Lifecycle?

Click to reveal 💡

3 3. Why do we need SQL when we already have Excel?

Click to reveal 💡

4 4. What is the main strength of Power BI vs Tableau?

Click to reveal 💡

5 5. Which Python library is used for manipulating DataFrames?

Click to reveal 💡

— Practical Research Assignment

# Assignment — Mini Research Task

Choose a company and analyze how it uses data.

 SELECT A COMPANY FOR YOUR CASE STUDY ASSIGNMENT:

Netflix

Amazon

Uber

Spotify

SUBMISSION REQUIREMENTS

1-Page Summary Report

## The 4 Questions to Answer for (Netflix):

- 1 What specific data types does this company collect daily?
- 2 How does the company transform raw data into actionable insights?
- 3 What critical business decisions are driven by their analytics models?
- 4 Which tools from our diploma roadmap (SQL, Python, Power BI) fit best?

Due Date: Before Session 02 for peer discussion

Practical Task



Netflix

Streaming & Entertainment

 DATA COLLECTED:

Watch history timestamps, pause/resume points, search queries, completion rates, and device telemetry.

 DRIVEN BUSINESS DECISIONS:

Personalized thumbnail generation, content recommendation engine, and budgeting \$17B original shows.

Reference Case

Netflix Case

iNSTANT

✓ SESSION 01 COMPLETED SUCCESSFULLY

# THANK YOU

Thank you for your active participation! Get ready for hands-on Excel data mastery in the next session.

[↺ Restart Presentation](#)[Open Session 02: Excel Basics →](#)